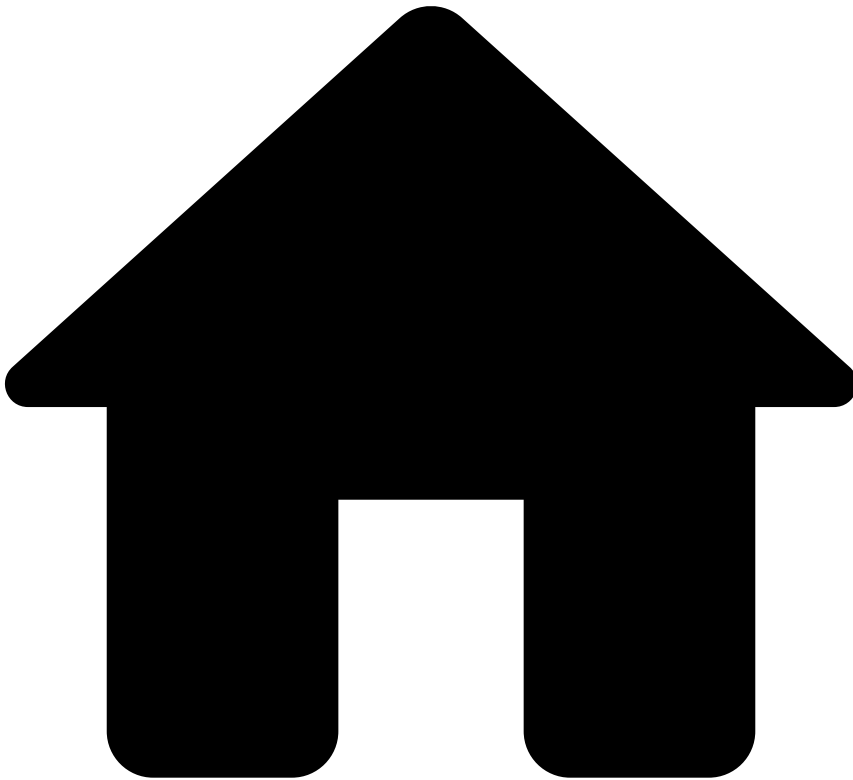


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Unusual Activity Rules

Description🔍📄

Contains rules that look for suspicious activity that is not yet associated with a particular fraud scenario but requires close monitoring. Subsequent transactions performed by alerted merchants may evolve into a specific fraud type such as bankruptcy, fraud or money laundering.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Score-Deviation-Weekly-UAR1

Comment

Average score from last week deviates by at least n standard deviation

Actions

- [Score-Deviation-Weekly-UAR1]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["weekly_fraud_score_limit"].threshold ??
default_threshold_list["weekly_fraud_score_limit"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

max_deviation = (threshold * event.std_score_1w_1w_dlay) + event.avg_score_1w_1w_dlay; // (n *
std dev of previous week fraud score) + avg score of previous week
```

Condition

```
event.avg_score_1w >= max_deviation // Average fraud score of last week // aggregated model score
per merchant from TF channel
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) current weekly average score of {event.avg_score_1w} deviates by at least {localvar.threshold} standard deviations compared to previous weekly average score of {event.avg_score_1w_1w_dlay}.

Mark As

N/A

Score-Deviation-Monthly-UAR2

Comment

Average score from last month deviates by at least n standard deviation

Actions

- [Score-Deviation-Monthly-UAR2]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["monthly_fraud_score_limit"].threshold ??
default_threshold_list["monthly_fraud_score_limit"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
```

```
default_threshold_list["min_trx_count"].threshold;

max_deviation = (threshold * event.std_score_4w_4w_dlay) + event.avg_score_4w_4w_dlay; // (n *
std dev of previous week fraud score) + avg score of previous week
```

Condition

```
event.avg_score_4w >= max_deviation // Average fraud score of last month // aggregated model
score per merchant from TF channel
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) current monthly average score of {event.avg_score_4w} deviates by at least {localvar.threshold} standard deviations compared to previous monthly average score of {event.avg_score_4w_4w_dlay}.

Mark As

N/A

MID-Track-List-UAR3

Comment

Merchant ID is in a track list

Actions

- Alert
- [MID-Track-List-UAR3]

Variables

```
listed = track_list.contains(event.merchant_id??");
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
listed
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) is in a track list.

Mark As

N/A

NDX-Limit-UAR4

Comment

NDX (non-delivery exposure) limit is equal to or above a set threshold

Actions

- [NDX-Limit-UAR4]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["ndx_limit"].threshold ??
default_threshold_list["ndx_limit"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

// For Rule Explanation
formatted_threshold = threshold / 100;
formatted_ndx = event.ndx_approx / 100;
```

Condition

```
event.ndx_approx >= threshold // if Merchant ADD == x days then NDX Apprx = Merchant Sum Amt x
days
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) Non Delivery Exposure is {localvar.formated_ndx} and is equal/above the limit of {localvar.formated_threshold} .

Mark As

N/A

ADD-Limit-UAR5

Comment

ADD (average delivery days) limit is equal to or above a set threshold

Actions

- [ADD-Limit-UAR5]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["add_limit"].threshold ??
default_threshold_list["add_limit"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
event.merchant_add >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) Average Delivery Days is {event.merchant_add} and is equal/above the hard limit of {localvar.threshold}.

Mark As

N/A

Distinct-Cards-Increase-UAR6

Comment

Number of distinct cards in 1 month has grown abnormally compared to the last 3 months

Actions

- [Distinct-Cards-Increase-UAR6]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["distinct_cards_1m_3m"].threshold ??
default_threshold_list["distinct_cards_1m_3m"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
// Number of distinct cards one month over the number of distinct of three months (delayed over
4w)
event.rt_dist_card_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) distinct cards number of one month is {event.rt_dist_card_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Dist-Country-Increase-UAR7

Comment

Number of distinct card countries in 1 month has grown abnormally compared to the last 3 months

Actions

- Alert
- [Dist-Country-Increase-UAR7]

Variables

```
threshold = multi_tenanted_threshold_list["distinct_cardcountry_1m_3m"].threshold ??
default_threshold_list["distinct_cardcountry_1m_3m"].threshold;
```

```
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
// Number of distinct card country one month over the number of distinct card country of three
months (delayed over 4w)
event.rt_dist_cardcntry_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) distinct card country number of one month is
{event.rt_dist_cardcntry_1m_to_3m}x higher than three months and is equal/above the limit of
{localvar.threshold}.

Mark As

N/A

Card-Manual-Increase-UAR8

Comment

Card manually entered payments rate of 1 month over 3 months is equal to or above a set threshold

Actions

- [Card-Manual-Increase-UAR8]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["card_manual_count_1m_3m_list"].threshold ??
default_threshold_list["card_manual_count_1m_3m_list"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
// Card manual payments rate of one month over the card manual rate of three months (delayed over
4w)
// Filters: card_entry_mode == 'manual'
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rtcomp_manual_cnt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) card manually entered payments rate of one month is
{event.rtcomp_manual_cnt_1m_to_3m}x higher than three months and is equal/above the limit of
{localvar.threshold}.

Mark As

N/A

Monthly-Count-Increase-UAR9

Comment

Number of payments in 1 month has grown abnormally compared to the last 3 months

Actions

- Alert
- [Monthly-Count-Increase-UAR9]

Variables

```
threshold = multi_tenanted_threshold_list["count_increase_1m_3m"].threshold ??
default_threshold_list["count_increase_1m_3m"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
// total count of approved payments one month over the total amount of approved payments of
three months (delayed over 4w)
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rt_cnt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) number of payments one month is {event.rt_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Monthly-Amt-Increase-UAR10

Comment

Total amount of payments in 1 month has grown abnormally compared to the last 3 months

Actions

- [Monthly-Amt-Increase-UAR10]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["amount_increase_1m_3m"].threshold ??
default_threshold_list["amount_increase_1m_3m"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
// total amount of approved payments one month over the total amount of approved payments of
three months (delayed over 4w)
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rt_amt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) total amount of payments one month is {event.rt_amt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Daily-Amount-Increase-UAR11

Comment

Total amount of payments in 1 day has grown abnormally compared to the last week

Actions

- [Daily-Amount-Increase-UAR11]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["amount_increase_1d_1w"].threshold ??
default_threshold_list["amount_increase_1d_1w"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
// total amount of approved payments one day over the total amount of approved payments of 1w
(delayed over 1day)
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rt_amt_1d_to_1w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) total amount of payments one day is {event.rt_amt_1d_to_1w}x higher than one week and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Peer-Avg-Ticket-Size-UAR12

Comment

Merchant daily average transaction value deviates by at least n Standard Deviations of the MCC 90 day average transaction value

Actions

- Alert
- [Peer-Avg-Ticket-Size-UAR12]

Variables

```
threshold = multi_tenanted_threshold_list["mcc_avg_ticket_size_limit"].threshold ??
default_threshold_list["mcc_avg_ticket_size_limit"].threshold;
threshold_count = multi_tenanted_threshold_list["min_count_succ_90d"].threshold ??
default_threshold_list["min_count_succ_90d"].threshold;

max_deviation = (threshold * event.mcc_std_accpt_amt_90d_1d_dlay) +
event.mcc_avg_accpt_amt_90d_1d_dlay; // (n * std dev of MCC succ trx amount last 90 days) + avg
MCC succ trx amount of last 90 days
```

Condition

```
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.avg_accpt_amt_1d >= max_deviation // Average succ trx amount last day >= Max Deviation
and event.cnt_accpt_90d_1d_dlay >= threshold_count // min 1 succ transaction per day
```

Explanation

Merchant ({event.merchant_id}) daily average ticket size is {event.avg_accpt_amt_1d} and deviates by at least {localvar.threshold} standard deviation compared to previous 90 days MCC average ticket size of {event.mcc_avg_accpt_amt_90d_1d_dlay}.

Mark As

N/A

Merch-Avg-Ticket-Size-UAR13

Comment

Merchant daily average transaction value deviates by at least n Standard Deviations of the Merchant 90 day average transaction value

Actions

- Alert
- [Merch-Avg-Ticket-Size-UAR13]

Variables

```
threshold = multi_tenanted_threshold_list["merch_avg_ticket_size_limit"].threshold ??
default_threshold_list["merch_avg_ticket_size_limit"].threshold;
threshold_count = multi_tenanted_threshold_list["min_count_succ_90d"].threshold ??
default_threshold_list["min_count_succ_90d"].threshold;

max_deviation = (threshold * event.std_accpt_amt_90d_1d_dlay) +
event.avg_accpt_amt_90d_1d_dlay; // (n * std dev of succ trx amount last 90 days) + avg succ trx
```

amount of last 90 days

Condition

```
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.avg_accpt_amt_1d >= max_deviation // Average succ trx amount last day >= Max Deviation
and event.cnt_accpt_90d_1d_dlay >= threshold_count // min 1 succ transaction per day
```

Explanation

Merchant ({event.merchant_id}) daily average ticket size is {event.avg_accpt_amt_1d} and deviates by at least {localvar.threshold} standard deviation compared to previous 90 days average ticket size of {event.avg_accpt_amt_90d_1d_dlay}.

Mark As

N/A

Gift-Prepaid-Card-Rate-UAR14

Comment

Daily gift or prepaid card payments rate is equal or above a set threshold

Actions

- [Gift-Prepaid-Card-Rate-UAR14]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["gift_prepaid_card_limit"].threshold ??
default_threshold_list["gift_prepaid_card_limit"].threshold;
```

```
//For Rule Explanation
formatted_threshold = threshold * 100;
formatted_rate = event.rt_prepaid_card_cnt_1d * 100;
```

Condition

```
// Approved Gift or prepaid payments rate of one day
// Filters: payment_auth_status == 'accepted' and payment_card_type == 'giftcard' or 'prepaid'
event.rt_prepaid_card_cnt_1d >= threshold
```

Explanation

Merchant ({event.merchant_id}) gift/prepaid card payments rate of one day is {localvar.formated_rate} and it's equal/above the limit of {localvar.formated_threshold}

Mark As

N/A